

Scores as of 1/27/19

Stock At A Glance

Micron Technology Inc. (NASDAQ: MU)

My Analysis: [Analyze This Stock](#)

All dollar values are displayed in US dollars.

Rule #1 Score				
Meaning	Moat	Mgmt	Predictability	Rule #1
91	100	94	97	95

[Add To Stock Watch](#)
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[Stock At a Glance](#) [Rule One Numbers](#) [Price Chart](#) [Margin of Safety](#)

What is the Town Analysis?

The Town Analysis gives you information that is necessary in order for you to make an informed valuation decision. Evaluate the numbers view and review the most salient financial numbers and growth rates for the last ten years. The chart view gives you a graphical representation of the numbers so you can see if this company has had perceptible and consistent growth and good returns on capital. Finally, the Town valuation view gives you tools necessary to determine what you think this company is worth.

Meaning	Rating
Relative Moat Score	100
Relative Rank	82
Rule One Meaning Score	91

Moat: Compound Growth Rate	10 Years	7 Years	5 Years	3 Years	1 Year	Rating
BVPS+Dividend Growth Rate	18.0%	21.1%	28.7%	37.9%	73.4%	100
EPS Growth	31.1%	88.5%	64.0%	69.6%	177.8%	100
OCPS Growth	32.8%	32.1%	57.2%	49.5%	113.4%	100
Sales Growth	17.9%	19.4%	27.4%	23.4%	49.5%	100
Rule One Moat Score						100

Management: Average Rate of Return	10 Years	7 Years	5 Years	3 Years	1 Year	Rating
ROE	10.7%	17.2%	24.1%	22.9%	43.8%	100
ROIC	8.5%	12.8%	18.1%	18.6%	39.2%	88
Debt	0.27 years earnings					97
Rule One Management Score						94

Business Description:

Micron Technology Inc along with its subsidiaries provide memory and storage solutions. Its product portfolio consists of memory and storage technologies such as DRAM, NAND, NOR and 3D XPoint memory.

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Details

CEO:	Sanjay Mehrotra		
Employees:	36,000	Market Cap:	\$43.68 Bil.
Issue Type:	CS		
Auditor:	PricewaterhouseCoopers LLP		
Last Audit:	UQ		
Indices:	Dow Jones U.S. Semiconductors Index (^DJUSSC), Semiconductor Memory (^QM31169146), S&P 500 (^SPX)		

Industry Classifications

Sector:	Technology	CIK:	723125
Industry:	Semiconductors		
Sub-Industry:	Semiconductor Memory		
NAICS:	Semiconductor and Related Device Manufacturing(334413)		
SIC:	Semiconductors And Related Devices (3674)		

Meaning**ITEM 1. BUSINESS****Overview**

Micron Technology, Inc., including its consolidated subsidiaries, is an industry leader in innovative memory and storage solutions. Through our global brands – Micron®, Crucial®, and Ballistix® – our broad portfolio of high-performance memory and storage technologies, including DRAM, NAND, NOR Flash and 3D XPoint memory, is transforming how the world uses information to enrich life. Backed by 40 years of technology leadership, our memory and storage solutions enable disruptive trends, including artificial intelligence, machine learning, and autonomous vehicles, in key market segments like cloud, data center, networking, and mobile.

We manufacture our products at our worldwide, wholly-owned and joint venture facilities. In recent years, we have increased our manufacturing scale and product diversity through strategic acquisitions, expansion, and various partnering arrangements.

We make significant investments to develop proprietary product and process technology, which is implemented in our manufacturing facilities. We generally increase the density per wafer and reduce manufacturing costs of each generation of product through advancements in product and process technology, such as our leading-edge line-width process technology and 3D NAND architecture. We continue to introduce new generations of products that offer improved performance characteristics, including higher data transfer rates, reduced package size, lower power consumption, improved read/write reliability, and increased memory density. Storage products incorporating NAND, a controller, and firmware constitute a significant and increasing portion of our sales. We generally develop firmware and expect to introduce proprietary controllers into our SSDs in the first half of 2019. Development of advanced technologies enables us to diversify our product portfolio toward a richer mix of differentiated, high-value solutions and to target high-growth markets.

We market our products through our internal sales force, independent sales representatives, distributors, and e-tailers, primarily to original equipment manufacturers and retailers located around the world. We face intense competition in the semiconductor memory and storage markets and, in order to remain competitive, we must continuously develop and implement new products and technologies and decrease manufacturing costs. Our success is largely dependent on market acceptance of our diversified portfolio of semiconductor-based memory and storage solutions, efficient utilization of our manufacturing infrastructure, successful ongoing development and integration of advanced product and process technology, return-driven capital spending, and successful R&D investments.

Products

Our product portfolio of memory and storage solutions, advanced solutions, and storage platforms are based on our high-performance semiconductor memory and storage technologies, including DRAM, NAND, 3D XPoint memory, and other technologies. We sell our products into various markets through our four business units (which are also our reportable segments) in various forms, including wafers, components, modules, SSDs and in MCPs that combine DRAM, NAND, and/or NOR with a controller and firmware. We are relentlessly focused on evolving our product portfolio to a richer mix of high-value solutions and cultivating deeper relationships with customers. Our position as a developer and manufacturer of DRAM, NAND, NOR and other emerging memory technologies uniquely enables us to collaborate with our customers to ensure our technology and engineering roadmaps deliver critical features. We continuously introduce new products on our advanced technologies, delivering performance, quality, and cost advantages to our customers.

Compute and Networking Business Unit

CNBU includes memory products and solutions sold into cloud server, enterprise, client, graphics, and networking markets. CNBU reported revenue of \$15.25 billion in 2018, \$8.62 billion in 2017, and \$4.53 billion in 2016. In 2018, we significantly increased our production of DRAM using 1Xnm technology and continued to focus on developing our 1Ynm technology. In 2018, we achieved volume production of our 8Gb GDDR6 memory, which delivers significant performance improvements over our GDDR5 design, and enables bandwidth-intensive applications in our core CNBU markets in a variety of applications such as artificial intelligence and networking.

Cloud Server: The cloud server market was CNBU's fastest growing market in 2018, particularly in datacenters, with significant increases in DRAM content per server. The cloud server market has been driven, in part, by intelligent edge devices capable of artificial intelligence and augmented reality that store and access data in the cloud. Artificial intelligence servers require significantly increasing quantities of DRAM and as the number and capabilities of these intelligent edge devices increase, more data is stored, processed, and accessed in the cloud, creating a virtuous cycle between the cloud and edge devices. We anticipate continued growth of our 1Xnm portfolio with the continued ramp of our second-generation 1Xnm 8Gb DDR4 products, which were validated with key partners and customers in 2018.

Enterprise: Similar to the cloud server market, the enterprise market is experiencing strong demand growth from intelligent edge devices that require rapid data analysis and storage in enterprise and cloud servers to enable machine learning, training, and inferencing. Our enterprise RDIMM DRAM memory modules provide the high performance, reliability, and integrity requirements for such applications. In 2018, we qualified our 32GB non-volatile module ("NVDIMM") at key OEMs and also began shipping in volume our 128GB through-silicon via-based ("TSV") RDIMMS.

Client: In 2018, we achieved significant production and sales to the client market from our 1Xnm technology. Our products sold to the client market support both PC unit growth, driven primarily by corporate replacement

cycles from upgraded operating systems, as well as increases in content per unit. Additionally, our products sold to the client market are incorporated into gaming and ultra-thin notebooks.

Graphics: Our GDDR5/5x DRAM graphics products are incorporated into applications providing virtual reality, augmented reality, and crypto-mining technology. In 2018, we benefitted from strong demand for graphics memory in gaming console applications, as well as a higher attach-rate of graphics DRAM products in performance and enthusiast graphics cards. In 2018, we migrated and scaled production of our 8Gb GDDR5 to our 1Xnm DRAM technology, which augmented production of our GDDR5/5x DRAM memory on our 20nm line-width technology. We remained focused on execution of technology transitions and achieved volume production of our 8Gb GDDR6 DRAM for the graphics and crypto-mining markets in 2018.

Networking: The networking memory market is characterized by long life-cycle DRAM products, and accordingly, a significant portion of our sales to the networking market consisted of products manufactured on our legacy 30nm and 25nm-series DRAM technology. In 2018, we accelerated a shift from DDR3 to DDR4 DRAM and began sales of 4Gb DDR4 DRAM into emerging 5G applications.

Mobile Business Unit

MBU includes memory products sold into smartphone and other mobile-device markets and includes discrete DRAM, discrete NAND, and managed NAND. MBU managed NAND includes eMMC and universal flash storage ("UFS") solutions, which each combine high-capacity NAND with a high-speed controller and firmware in a small ball-grid array, and eMCP products, which combine an eMMC/UFS solution with LPDRAM. MBU reported revenue of \$6.58 billion in 2018, \$4.42 billion in 2017, and \$2.57 billion in 2016. In 2018, we announced new 64-layer, second-generation 3D NAND storage products, which support the high-speed UFS 2.1 standard and eMMC 5.1 standard. These new mobile solutions are based on our industry-leading TLC 3D NAND technology, empowering smartphone makers to enhance the user experience with next-generation mobile features such as artificial intelligence, virtual reality, and facial recognition. Our 1Xnm LPDRAM solutions provide power efficiency, particularly critical to our mobile customers, and our 1Ynm 12Gb LPDDR4 solutions, the highest capacity LPDRAM monolithic die available in the industry, provide both power efficiency and higher capacity to our mobile customers.

Smartphone: In 2018, we achieved product qualification of our 1Xnm LPDDR4 DRAM with major mobile phone OEMs. Our LPDRAM offers low-power, high-performance solutions to perform in extreme environments demanded by high-end smartphones. High-end smartphones incorporate higher levels of NAND and LPDRAM that enable features such as larger 4K displays, multiple high-resolution cameras, and 4K high-dynamic range video recording. Additionally, our smartphone products are utilized by OEMs to enable artificial intelligence, augmented reality, and life-like virtual reality capabilities into high-end phones, including facial and voice recognition, real-time translation, fast image search, and scene detection. In 2018, our managed NAND products achieved strong growth, including our new 128GB NAND plus 4GB DRAM MCP and our first high-performance UFS managed NAND products introduced in the fourth quarter of 2018.

Storage Business Unit

SBU includes SSDs and component-level solutions sold into enterprise and cloud, client, and consumer storage markets as well as other discrete storage products sold in component and wafer forms to the removable storage markets. SBU sales also include "non-trade" products consisting of products manufactured and sold to Intel through IMFT under a long-term supply agreement at prices approximating cost, which included 3D XPoint memory and NAND products. SBU reported revenue of \$5.02 billion in 2018, \$4.51 billion in 2017, and \$3.26 billion in 2016. In 2018, we continued to ramp our 64-layer 3D NAND technology and achieved bit output crossover relative to 32-layer in the second half of 2018. In 2018, we also extended our leadership position in 3D

NAND technology by delivering the industry's first commercially available QLC 3D NAND technology. Leveraging our 64-layer structure, the new QLC NAND technology achieves 1 terabit ("Tb") density per die, which has a 33% higher array density as compared to TLC, enabling new operating points for density and cost in the enterprise, cloud, and client-storage markets. In 2018, we advanced development of our third-generation 96-tier 3D NAND structure, providing a 50 percent increase in layers. Both the 64-layer QLC and 96-layer TLC 3D NAND technologies utilize CMOS under the array ("CuA") technology to reduce die sizes and deliver improved performance when compared to competitive approaches. By leveraging four planes versus two, our new NAND flash memory can write and read more cells in parallel, which delivers faster throughput and higher bandwidth at the system level.

SSDs: SSD storage products incorporate NAND, a controller, and firmware and offer benefits over HDDs of a smaller form factor, faster read and write speeds, and solid-state architecture. SSDs offer significant performance and features, including speed, reliability, and lower power consumption. We offer SSD solutions utilizing our NAND technology to the enterprise and cloud, client, and consumer markets.

Enterprise and Cloud SSDs: SBU sales to the enterprise and cloud SSD markets in 2018 consisted primarily of our flagship SATA 5100 and 5200 series SSDs. In 2018, our SATA 5200 series SSD achieved qualification at enterprise server OEMs, cloud service providers, and enterprise customers. Similar to trends in the memory market, the enterprise and cloud storage markets have been driven by intelligent edge devices capable of artificial intelligence, augmented reality, and other features that store, access, and analyze data in the cloud. Artificial intelligence servers require significantly higher SSD capacity, and our 64-layer QLC NAND technology provides cost-optimized storage solutions, providing significantly lower total cost of ownership for read-intensive cloud workloads. Our 5200 series SATA SSDs, which deliver best-in-class performance and capacity, are based on the same proven architecture as our 5100 series. We shipped our first 5200 series SATA SSDs in the third quarter of 2018 and received broad acceptance in the enterprise and cloud SSD markets. By leveraging our advanced CuA NAND in enterprise and cloud SSDs, we deliver low cost, high density, high performance storage solutions.

Client SSDs: SBU sales to the client SSD market in 2018 consisted primarily of our 1100 series 3D NAND SATA Client SSD, which is targeted for leading personal computer OEMs as a replacement to HDDs. Our client SSDs, used in notebooks, desktops, workstations, and related consumer applications, deliver high performance, power efficiency, security, and capacity to our customers. In the first half of 2019, we expect to introduce our 2200 series 3D NAND PCIe client SSD incorporating our internally-developed controller, enabling us to offer additional differentiated storage solutions for our client customers.

Consumer SSDs: SBU sales to the consumer SSD market in 2018 consisted primarily of our Crucial-branded MX500 SATA SSD, utilizing our 64-layer TLC 3D NAND. Similar to the client SSD market, our consumer SSD solutions are replacing HDDs as end-users seek the higher performance, power savings, and reliability of our SSDs.

Components and Wafers: SBU sales of components and wafers in 2018 consisted primarily of our 32-layer TLC NAND technology and our 64-layer TLC and QLC NAND technology. We continue to transition our business from a storage components supplier to a storage solutions provider with a richer mix of high-value solutions such as SSDs and mobile managed NAND. As a result, SBU sales of products in component and wafer form declined in 2018 as compared to 2017.

3D XPoint memory: 3D XPoint memory has 10 times the chip density of DRAM, 1,000 times the endurance capability of NAND, and is 1,000 times faster than NAND. These specifications create a significant value opportunity for 3D XPoint memory in solutions between DRAM and NAND in the memory and storage hierarchy. Trends in machine learning, big data analytics, and artificial intelligence are driving demand for the

features offered by 3D XPoint memory. We are collaborating with our customers to develop 3D XPoint memory products and expect to sample such products in late calendar 2019.

Embedded Business Unit

EBU includes memory and storage products sold into automotive, industrial, and consumer markets and includes discrete DRAM, discrete NAND, managed NAND, and NOR. EBU reported revenue of \$3.48 billion in 2018, \$2.70 billion in 2017, and \$1.94 billion in 2016. The embedded market is characterized by long life-cycle DRAM and NAND products manufactured on our mature process technologies. Our embedded products enable edge devices to store, connect, and share information in the growing internet of things ("IoT") and are utilized in a diverse set of applications in the automotive, industrial, and consumer markets.

Automotive: Our DDR3 DRAM and eMMC managed NAND automotive memory and storage products enable connected, large display infotainment systems and higher definition 4K displays and support improved voice and gesture control in automotive applications. Our comprehensive and expanding portfolio of DRAM, NAND, and NOR solutions to the automotive market, as well as our extensive customer support network, also support advancements in autonomous driving and automated driver assistance systems, which require high reliability and high performance memory and storage.

Industrial: Our industrial products, featuring SLC and MLC NAND, NOR, DDR3 DRAM, and MCP managed NAND, enable applications in the growing industrial IoT market, including factory automation, transportation, and surveillance. In 2018, we announced availability of our 128GB and 256GB density of edge storage microSD card solutions and collaboration with several leading video surveillance solution providers to promote surveillance-grade edge storage, utilizing our 64-layer TLC 3D NAND technology. This newly released solution enables greater capacity in a smaller space, delivering up to 30 days of surveillance video storage in the camera.

Consumer: Our DDR3 DRAM, SLC NAND, and eMCP managed NAND products sold into the consumer market are used in a diverse set of consumer products, including service provider and set-top boxes, digital still and video cameras, home networking, ultra-high definition televisions, and many more applications. Our embedded memory and storage solutions enable edge devices in the consumer products market to store, connect, and share information in the IoT.

Manufacturing

We manufacture our products at our worldwide, wholly-owned and joint venture facilities located in Taiwan, Singapore, the United States, Japan, and China and also utilize subcontractors to perform certain manufacturing processes. Nearly all of our products are manufactured on 300mm wafers in facilities that generally operate 24 hours per day, seven days per week. Semiconductor manufacturing is extremely capital intensive, requiring large investments in sophisticated facilities and equipment. A significant portion of our semiconductor equipment is generally replaced every five to seven years with increasingly advanced equipment. Our DRAM, NAND, 3D XPoint memory, and NOR Flash products share a number of common manufacturing processes, enabling us to leverage much of our product and process technology and manufacturing infrastructure across these product lines.

Our process for manufacturing semiconductor products is complex and involves a number of precise steps, including wafer fabrication, assembly, and test. Efficient production of semiconductor products requires utilization of advanced semiconductor manufacturing techniques and effective deployment of these techniques across multiple facilities. The primary determinants of manufacturing cost are process line-width, 3D non-volatile layers, NAND cell levels, process complexity, including number of mask layers and fabrication steps, and manufacturing yield. Other factors that contribute to manufacturing costs are the cost and sophistication of manufacturing equipment, equipment utilization, process complexity, cost of raw materials, labor productivity,

package type, cleanliness of our manufacturing environment, and utilization of subcontractors to perform certain manufacturing processes. We continuously enhance our production processes, increasing bits per wafer and transitioning to higher density products. In 2018, we significantly increased our volume production of 1Xnm process node DRAM and expect to achieve bit crossover by the end of the first quarter of 2019. In 2018, we continued to ramp our 64-layer 3D NAND technology and achieved bit output crossover relative to 32-layer in the second half of 2018.

Wafer fabrication occurs in a highly-controlled clean environment to minimize dust and other yield and quality-limiting contaminants. Despite stringent manufacturing controls, individual circuits may be nonfunctional or wafers may need to be scrapped due to equipment errors, minute impurities in materials, defects in photomasks, circuit design marginalities or defects, and air particle defects. Success of our manufacturing operations depends largely on minimizing defects to maximize yield of high-quality circuits. In this regard, we employ rigorous quality controls throughout the manufacturing, screening, and testing processes. We are able to recover certain devices by testing and grading them to their highest level of functionality.

We sell semiconductor products in both packaged and unpackaged (i.e., "bare die") forms. Our packaged products include memory modules, SSDs, and managed NAND including MCPs and eMMCs. We assemble many products in-house and, in some cases, outsource assembly services for certain memory modules, SSDs, and MCPs.

We test our products at various stages in the manufacturing process, conduct numerous quality control inspections throughout the entire production flow, and perform high temperature burn-in on finished products. In addition, we use our proprietary AMBYX™ line of intelligent test and burn-in systems to perform simultaneous circuit tests of semiconductor die during the burn-in process, capturing quality and reliability data and reducing testing time and cost.

In recent years, we have produced an increasingly broad portfolio of products and system solutions, which enhances our ability to allocate resources to our most profitable products but also increases the complexity of our manufacturing and supply chain operations. Although our product lines generally use similar manufacturing processes, our cost efficiency can be affected by frequent conversions to new products, the allocation of manufacturing capacity to more complex, smaller-volume products, and the reallocation of manufacturing capacity across various product lines.

Arrangements with Intel

IMFT

Since 2006, we have owned 51% of IMFT, a joint venture between us and Intel. IMFT is governed by a Board of Managers, for which the number of managers appointed by each member varies based on the members' respective ownership interests. IMFT manufactures semiconductor products exclusively for its members under a long-term supply agreement at prices approximating cost. In the first quarter of 2018, IMFT discontinued production of NAND and subsequent to that time has been entirely focused on 3D XPoint memory production. IMFT sales to Intel were \$507 million, \$438 million, and \$457 million in 2018, 2017, and 2016, respectively.

The IMFT joint venture agreement extends through 2024 and includes certain buy-sell rights. At any time through December 2018, Intel can put to us, and from January 2019 through December 2021, we can call from Intel, Intel's interest in IMFT, in either case, for a price that approximates Intel's interest in the net book value of IMFT plus member debt at the time of the closing. If Intel exercises its put right, we can elect to set the closing date of the transaction any time between six months and two years following such election by Intel and we can elect to receive financing of the purchase price from Intel for one to two years from the closing date. If we

exercise our call right, Intel can elect to set the closing date of the transaction to be any time between six months and one year following such election. Following the closing date resulting from exercise of either the put or the call, we will continue to supply to Intel for a period of one year between 50% and 100%, at Intel's choice, of Intel's immediately preceding six-month period pre-closing volumes of IMFT products for the first six-month period following the closing and between 0% and 100%, at Intel's choice, of Intel's first six-month period following the closing volumes of IMFT products for the second six-month period following the closing, at a margin that varies depending on whether the put or call was exercised.

IMFT's capital requirements are generally determined based on an annual plan approved by the members, and capital contributions to IMFT are requested as needed. Capital requests are made to the members in proportion to their then-current ownership interest. Members may elect to not contribute their proportional share, and in such event, the contributing member may elect to contribute any amount of the capital request, either in the form of an equity contribution or member debt financing. In 2018, Intel provided debt financing of \$1.01 billion to IMFT pursuant to the terms of the IMFT joint venture agreement. Under the supply agreement, the members have rights and obligations to the capacity of IMFT in proportion to their investment, including member debt financing. Any capital contribution or member debt financing results in a proportionate adjustment to the sharing of output on an eight-month lag. Members pay their proportionate share of fixed costs associated with IMFT's capacity.

R&D Arrangements

We have agreements to jointly develop NAND and 3D XPoint technologies with Intel. We continue to jointly develop NAND technologies with Intel through the third generation of 3D NAND, which is expected to be completed in the second half of 2019. In the second quarter of 2018, we and Intel agreed to independently develop subsequent generations of 3D NAND in order to better optimize the technology and products for our respective business needs. We continue to jointly develop 3D XPoint technologies with Intel through the second generation of 3D XPoint technology, which is expected to be completed in the second half of 2019. To better optimize 3D XPoint technology for our product roadmap and maximize the benefits for our customers and shareholders, in the fourth quarter of 2018, we announced that we will no longer jointly develop with Intel subsequent generations of 3D XPoint technology. As a result of the above actions, we expect reimbursements under our cost-sharing agreements to decrease in early fiscal 2019.

Supply Chain, Materials, and Use of Third-Party Service Providers

Our supply chain and operations are dependent on the availability of materials that meet exacting standards and the use of third parties to provide us with components and services. We generally have multiple sources of supply for our raw materials and services. However, only a limited number of suppliers are capable of delivering certain raw materials and services that meet our standards and, in some cases, materials, components, or services are provided by a single supplier. Various factors could reduce the availability of raw materials or components such as chemicals, silicon wafers, gases, photoresist, controllers, substrates, lead frames, printed circuit boards, targets, and reticle glass blanks. Shortages or increases in lead times may occur from time to time in the future. Our manufacturing processes are also dependent on our relationships with third-party manufacturers of controllers used in a number of our products and with outsourced semiconductor assembly and test providers, contract manufacturers, logistic carriers, and other service providers. Certain raw materials are primarily available in certain countries, including rare earth minerals available primarily from China, and trade disputes or other political or economic conditions may limit our availability to obtain such raw materials. We and/or our suppliers and service providers could be affected by tariffs, embargoes or other trade restrictions, as well as laws and regulations enacted in response to concerns regarding climate change, conflict minerals, and responsible sourcing practices, which could limit the supply of our raw materials and/or increase the cost. In

addition, disruptions in transportation lines could delay our receipt of raw materials. Lead times for the supply of raw materials have been extended in the past. The disruption of our supply of raw materials, components, services, or the extension of our lead times could have a material adverse effect on our business, results of operations, or financial condition. We monitor and manage supply-chain activities to mitigate risks associated with raw materials and service providers.

Marketing and Customers

We continue to transform how we interact with our customers from transactional opportunistic sales of standardized memory components to collaborative relationships where we work with our customers to understand their unique opportunities and challenges. We engage with our customers early in the product life-cycle to identify and design features and performance characteristics into our products that our customers need in their end products, and then manufacture products that better anticipate and fit their changing needs. By collaborating with our customers on their design needs in a changing end market, we differentiate our memory and storage solutions, which provides greater value to our customers.

Our semiconductor memory and storage products are offered under our Micron, Crucial, and Ballistix brand names and through private labels. We market our semiconductor memory and storage products primarily through our own direct sales force and maintain sales or representative offices in our primary markets around the world. We sell our Crucial-branded products through a web-based customer direct sales channel as well as through channel and distribution partners. Our products are also offered through independent sales representatives, distributors, and e-tailers. Our independent sales representatives obtain orders subject to final acceptance by us, and we make shipments against the orders directly to our customers. Our distributors carry our products in inventory and typically sell a variety of other semiconductor products, including competitors' products. We maintain inventory at locations in close proximity to certain key customers to facilitate rapid delivery of products. Many of our customers require a thorough review or qualification of semiconductor products, which may take several months.

In each of the last three years, approximately one-half of our total net sales were to our top ten customers. For other information regarding our concentrations and customers, see "Part II – Item 8. Financial Statements and Supplementary Data – Notes to Consolidated Financial Statements – Certain Concentrations."

Backlog

Because of volatile industry conditions, customers are generally reluctant to enter into long-term, fixed-price contracts. Accordingly, new order volumes for our memory and storage products may fluctuate significantly. We typically accept orders with acknowledgment that the terms may be adjusted to reflect market conditions at the date of shipment. For these reasons, we do not believe that our order backlog as of any particular date is a reliable indicator of actual sales for any succeeding period.

Product Warranty

Because the design and manufacturing process for semiconductor products is highly complex, it is possible that we may produce products that do not comply with applicable specifications, contain defects, or are otherwise incompatible with end uses. In accordance with industry practice, we generally provide a limited warranty that our products are in compliance with applicable specifications existing at the time of delivery and will operate to those specifications during a stated warranty period. Under our standard terms and conditions of sale, liability for certain failures of product during a stated warranty period is usually limited to repair or replacement of defective items or return of, or a credit with respect to, amounts paid for such items. Under

certain circumstances, we provide more extensive limited warranty coverage than that provided under our standard terms and conditions.

Competition

We face intense competition in the semiconductor memory and storage markets from a number of companies, including Intel; Samsung Electronics Co., Ltd.; SK Hynix Inc.; Toshiba Memory Corporation; and Western Digital Corporation. Some of our competitors are large corporations or conglomerates that may have greater resources to invest in technology, capitalize on growth opportunities, and withstand downturns in the semiconductor markets in which we compete. Consolidation of industry competitors could put us at a competitive disadvantage. In addition, some governments have provided and may continue to provide significant assistance, financial or otherwise, to some of our competitors or to new entrants and may intervene in support of national industries and/or competitors. In particular, we face the threat of increasing competition as a result of significant investment in the semiconductor industry by the Chinese government and various state-owned or affiliated entities that is intended to advance China's stated national policy objectives. In addition, the Chinese government may restrict us from participating in the China market or may prevent us from competing effectively with Chinese companies.

Our competitors generally seek to increase silicon capacity, improve yields, and reduce die size in their product designs which may result in significant increases in worldwide supply and downward pressure on prices. Increases in worldwide supply of semiconductor memory and storage also result from fabrication capacity expansions, either by way of new facilities, increased capacity utilization, or reallocation of other semiconductor production to semiconductor memory and storage production. Our competitors may increase capital expenditures resulting in future increases in worldwide supply. We and some of our competitors have plans to ramp, or are constructing or ramping, production at new fabrication facilities. Increases in worldwide supply of semiconductor memory and storage, if not accompanied by commensurate increases in demand, would lead to declines in average selling prices for our products and would adversely affect our business, results of operations, and financial condition. If competitors are more successful at developing or implementing new product or process technology, their products could have cost or performance advantages.

Certain of our memory and storage products are manufactured to industry standard specifications and, as such, have similar performance characteristics to those of our competitors. For these products, the principal competitive factors are generally price and performance characteristics including: operating speed, power consumption, reliability, compatibility, size, and form factors.

Research and Development

Our process technology R&D efforts are focused primarily on development of process technology that enables continuous improvement to cost structures and performance enhancements for our future products. We are also focused on developing new fundamentally different memory structures, materials, and packages, which are designed to facilitate our transition to next generation products. Additional process technology R&D efforts focus on the enablement of advanced computing, storage, and mobile memory architectures, the investigation of new opportunities that leverage our core semiconductor expertise, and the development of new manufacturing materials. Product design and development efforts include our high density DDR4 and DDR5 DRAM and LPDRAM products as well as high density and mobile 3D NAND (including TLC and QLC technologies), 3D XPoint memory, SSDs (including firmware and controllers), managed NAND, specialty memory, and other memory technologies and systems.

To compete in the semiconductor memory and storage markets, we must continue to develop technologically advanced products and processes. We believe that expansion of our semiconductor product

offerings is necessary to meet expected market demand for specific memory and storage products and solutions. Our process, design, and package development efforts occur at multiple locations across the world, with our largest R&D center located in Boise, Idaho and other R&D centers in Japan, China, Italy, Singapore, Taiwan, and other sites in the United States.

R&D expenses vary primarily with the number of development wafers processed, the cost of advanced equipment dedicated to new product and process development, and personnel costs. Because of the lead times necessary to manufacture our products, we typically begin to process wafers before completion of performance and reliability testing. Development of a product is deemed complete when it is qualified through reviews and tests for performance and reliability. R&D expenses can vary significantly depending on the timing of product qualification.

Our R&D expenses were \$2.14 billion, \$1.82 billion, and \$1.62 billion for 2018, 2017, and 2016, respectively. We share the cost of certain product and process development activities under development agreements with partners, including agreements to jointly develop NAND and 3D XPoint technologies with Intel. These R&D expenses reflect net reductions of \$201 million, \$213 million, and \$205 million for 2018, 2017, and 2016, respectively, as a result of reimbursements under our cost-sharing arrangements with development partners.

Geographic Information

See "Part II – Item 8. Financial Statements and Supplementary Data – Notes to Consolidated Financial Statements – Geographic Information."

Patents and Licenses

We are a recognized leader in per capita and quality of patents issued. As of August 30, 2018, we owned approximately 13,750 active U.S. patents and 5,000 active foreign patents. In addition, we have thousands of U.S. and foreign patent applications pending. Our patents have various terms expiring through 2038.

From time to time, we sell and/or license our technology to other parties and continue to pursue opportunities to monetize our investment in our intellectual property through partnering and other arrangements. We have also jointly developed memory and storage product and process technology with third parties on a limited basis.

We have a number of patent and intellectual property license agreements and have, from time to time, licensed or sold our intellectual property to third parties. Some of these license agreements require us to make one-time or periodic payments while others have resulted in us receiving payments. We may need to obtain additional licenses or renew existing license agreements in the future, and we may enter into additional sales or licenses of intellectual property and partnering arrangements. We are unable to predict whether these license agreements can be obtained or renewed on terms acceptable to us.

Employees

As of August 30, 2018, we had approximately 36,000 employees.

Environmental Compliance

We approach environmental stewardship and sustainability proactively to ensure we meet all government regulations regarding raw materials, discharges, emissions, and solid wastes from our manufacturing processes. Our wafer fabrication facilities continued to conform to the requirements of the International Organization for Standardization ("ISO") 14001 environmental management systems standard to ensure we are continuously improving our performance. As part of the ISO 14001 framework, we meet requirements in

environmental policy, compliance, planning, management, structure and responsibility, training, communication, document control, operational control, emergency preparedness and response, record keeping, and management review. While we have not experienced any material adverse effects to our operations from environmental regulations, changes in the regulations could necessitate additional capital expenditures, modification of our operations, or other compliance actions.

Directors and Executive Officers of the Registrant

Our executive officers are appointed annually by our Board of Directors (the "Board") and our directors are elected annually by our shareholders. Any directors appointed by the Board to fill vacancies on the Board serve until the next election by our shareholders. All officers and directors serve until their successors are duly chosen or elected and qualified, except in the case of earlier death, resignation, or removal.

As of August 30, 2018, the following executive officers and directors were subject to the reporting requirements of Section 16(a) of the Securities Exchange Act of 1934, as amended.

Name	Age	Officer/ Director	Position
		Since	
April S. Arnzen	47	2015	Senior Vice President, Human Resources
Manish Bhatia	46	2018	Executive Vice President, Global Operations
Scott J. DeBoer	52	2007	Executive Vice President, Technology Development
Sanjay Mehrotra	60	2017	President and Chief Executive Officer, Director
Joel L. Poppen	54	2013	Senior Vice President, Legal Affairs, General Counsel, and Corporate Secretary
Sumit Sadana	49	2017	Executive Vice President and Chief Business Officer
Steven L. Thorsen, Jr.	53	2012	Senior Vice President, Worldwide Sales
David A. Zinsner	49	2018	Senior Vice President and Chief Financial Officer
Robert L. Bailey	61	2007	Director
Richard M. Beyer	69	2013	Director
Patrick J. Byrne	57	2011	Director
Mercedes Johnson	64	2005	Director
Lawrence N. Mondry	58	2005	Director
Robert E. Switz	71	2006	Chairman of the Board of Directors

April S. Arnzen joined us in December 1996 and has served in various leadership positions since that time. Ms. Arnzen was named Senior Vice President, Human Resources in June 2017. Ms. Arnzen holds a BS in Human Resource Management and Marketing from the University of Idaho, and is a graduate of the Stanford Graduate School of Business Executive Program.

Manish Bhatia joined us in October 2017 as our Executive Vice President of Global Operations. From May 2016 to October 2017, Mr. Bhatia served as the Executive Vice President of Silicon Operations at Western Digital Corporation. From March 2010 to May 2016, Mr. Bhatia held several executive roles at SanDisk Corporation including Executive Vice President of Worldwide Operations when it was acquired by Western Digital in May 2016. Mr. Bhatia holds a BS and MS in Mechanical Engineering and an MBA, each from the Massachusetts Institute of Technology.

Scott J. DeBoer joined us in February 1995 and has served in various leadership positions since that time. Dr. DeBoer was named Executive Vice President, Technology Development in June 2017. Dr. DeBoer holds a PhD in Electrical Engineering and an MS in Physics from Iowa State University. He completed his undergraduate degree at Hastings College.

Sanjay Mehrotra joined us in May 2017 as our President, Chief Executive Officer, and Director. Mr. Mehrotra co-founded and led SanDisk Corporation as a start-up in 1988 until its eventual sale in May 2016, serving as its President and Chief Executive Officer from January 2011 to May 2016, and as a member of its Board of Directors from July 2010 to May 2016. Mr. Mehrotra served as a member of the Board of Directors for Cavium, Inc. from July 2009 until July 2018 and for Western Digital Corp. from May 2016 to February 2017. Mr. Mehrotra holds a BS and an MS in Electrical Engineering and Computer Science from the University of California, Berkeley and is a graduate of the Stanford Graduate School of Business Executive Program.

Joel L. Poppen joined us in October 1995 and has held various leadership positions since that time. Mr. Poppen was named Senior Vice President, Legal Affairs, General Counsel, and Corporate Secretary in June 2017. Mr. Poppen holds a BS in Electrical Engineering from the University of Illinois and a JD from the Duke University School of Law.

Sumit Sadana joined us in June 2017 as our Executive Vice President and Chief Business Officer. From April 2010 to May 2016, Mr. Sadana served in various roles at SanDisk Corporation, including Executive Vice President, Chief Strategy Officer, and General Manager, Enterprise Solutions when it was acquired by Western Digital in May 2016. Mr. Sadana currently serves on the Board of Directors of Silicon Laboratories, Inc. Mr. Sadana holds a B.Tech. in Electrical Engineering from the Indian Institute of Technology, Kharagpur, India and an MS in Electrical Engineering from Stanford University.

David A. Zinsner joined us in February 2018 as our Senior Vice President and Chief Financial Officer. From April 2017 to February 2018, Mr. Zinsner served as the President and Chief Operating Officer of Affirmed Networks. From January 2009 to April 2017, Mr. Zinsner served as the Senior Vice President of Finance and Chief Financial Officer of Analog Devices. From July 2005 to January 2009, Mr. Zinsner served as the Senior Vice President and Chief Financial Officer of Intersil Corporation. Mr. Zinsner holds an MBA, Finance and Accounting from Vanderbilt University and a BS in Industrial Management from Carnegie Mellon University.

Steven L. Thorsen, Jr. joined us in September 1988 and has served in various leadership positions since that time. Mr. Thorsen was named Senior Vice President, Worldwide Sales in June 2017. Mr. Thorsen holds a BA in Business Administration from Washington State University. On September 20, 2018, Mr. Thorsen announced his intention to retire from Micron in early November 2018. Mr. Thorsen served as our Senior Vice President, Worldwide Sales through September 30, 2018.

Robert L. Bailey was Chief Executive Officer of Blue Willow Systems, Inc. from August 2017 until August 2018. Blue Willow is a software as a service resident safety platform for senior living facilities. Mr. Bailey was the Chairman of the Board of Directors of PMC-Sierra, Inc. from 2005 until May 2011 and also served as PMC's Chairman from February 2000 until February 2003. Mr. Bailey served as a director of PMC from October 1996 to May 2011. He also served as the Chief Executive Officer of PMC from July 1997 until May 2008. Within the past five years, Mr. Bailey also served on the Board of Directors of Entropic Communications. Mr. Bailey holds a BS in Electrical Engineering from the University of Bridgeport and an MBA from the University of Dallas.

Richard M. Beyer was Chairman and Chief Executive Officer of Freescale Semiconductor, Inc. from 2008 through June 2012 and served as a director with Freescale until April 2013. Prior to Freescale, Mr. Beyer was President, Chief Executive Officer and a director of Intersil Corporation from 2002 to 2008. He also has previously served in executive management roles at FVC.com, VLSI Technology, and National Semiconductor Corporation.

Within the past five years, Mr. Beyer served on the Board of Directors of Microsemi Corporation, Analog Devices, Inc., and Freescale. He currently serves on the Board of Directors of Dialog Semiconductor. Mr. Beyer served three years as an officer in the United States Marine Corps. He holds a BA and an MA in Russian from Georgetown University and an MBA in Marketing and International Business from Columbia University Graduate School of Business. Mr. Beyer is the Chair of the Board of Directors' Governance and Sustainability Committee.

Patrick J. Byrne has served as Senior Vice President of Fortive Corporation since July 2016, when Danaher Corporation completed the separation of its Test & Measurement and Industrial Technologies segments. Mr. Byrne was President of Tektronix, a subsidiary of Danaher, from July 2014 to July 2016. Previously, he was Vice President of Strategy and Business Development and Chief Technical Officer of Danaher from November 2012 to July 2014. Danaher designs, manufactures, and markets innovative products and services to professional, medical, industrial, and commercial customers. Mr. Byrne served as Director, President and Chief Executive Officer of Intermec, Inc. from 2007 to May 2012. Within the past five years, Mr. Byrne served on the Board of Directors of Flow International. Mr. Byrne holds a BS in Electrical Engineering from the University of California, Berkeley and an MS in Electrical Engineering from Stanford University.

Mercedes Johnson was the Senior Vice President and Chief Financial Officer of Avago Technologies Limited, a supplier of analog interface components for communications, industrial, and consumer applications, from December 2005 to August 2008. She also served as the Senior Vice President, Finance of Lam Research Corporation from June 2004 to January 2005 and as Lam's Chief Financial Officer from May 1997 to May 2004. Ms. Johnson holds a degree in Accounting from the University of Buenos Aires and currently serves on the Board of Directors for Juniper Networks, Inc., Teradyne, Inc., and Synopsys, Inc. She also served on the Board of Directors for Intersil Corporation from August 2005 to February 2017. Ms. Johnson is the Chair of the Board of Directors' Audit Committee and Finance Committee.

Lawrence N. Mondry has been the President and Chief Executive Officer of Stream Gas & Electric, Ltd., a provider of energy, mobile, and protective services, since February 2016. Mr. Mondry was the Chief Executive Officer of Apollo Brands, a consumer products portfolio company, from February 2014 to February 2015. Mr. Mondry was the Chief Executive Officer of Flexi Compras Corporation, a rent-to-own retailer, from June 2013 to February 2014. Mr. Mondry was the President and Chief Executive Officer of CSK Auto Corporation, a specialty retailer of automotive aftermarket parts, from August 2007 to July 2008. Prior to his appointment at CSK, Mr. Mondry served as the Chief Executive Officer of CompUSA Inc. from November 2003 to May 2006. Mr. Mondry is the Chair of the Board of Directors' Compensation Committee.

Robert E. Switz was the Chairman, President, and Chief Executive Officer of ADC Telecommunications, Inc., a supplier of network infrastructure products and services, from August 2003 until December 2010, when Tyco Electronics Ltd. acquired ADC. Mr. Switz joined ADC in 1994 and throughout his career there held numerous leadership positions. Within the past five years, Mr. Switz served on the Board of Directors of GT Advanced Technologies Inc., Broadcom Corporation, Cyan, Inc., Pulse Electronics Corporation, Leap Wireless International, Inc., and Gigamon, Inc. Mr. Switz currently serves on the Board of Directors for Marvell Technology Group Ltd. and FireEye, Inc. Mr. Switz holds an MBA from the University of Bridgeport and a BS in Business Administration from Quinnipiac University. Mr. Switz was appointed Chairman of the Board of Directors in 2012.

There are no family relationships between any of our directors or executive officers.

Available Information

Micron, a Delaware corporation, was incorporated in 1978. Our executive offices are located at 8000 South Federal Way, Boise, Idaho 83716-9632 and our telephone number is (208) 368-4000. Information about us is available at our website, www.micron.com. Also available on our website are our: Corporate Governance

Guidelines, Governance and Sustainability Committee Charter, Compensation Committee Charter, Audit Committee Charter, Finance Committee Charter, and Code of Business Conduct and Ethics. Any amendments or waivers of our Code of Business Conduct and Ethics will also be posted on our website within four business days of the amendment or waiver. Copies of these documents are available to shareholders upon request. Information contained or referenced on our website is not incorporated by reference and does not form a part of this Annual Report on Form 10-K.

We use our investor relations website <http://investors.micron.com> as a routine channel for distribution of important information, including news releases, analyst presentations, and financial information. Our filings are available free of charge on our website as soon as reasonably practicable after they are electronically filed with, or furnished to, the U.S. Securities and Exchange Commission, including our annual and quarterly reports on Forms 10-K and 10-Q and current reports on Form 8-K, our proxy statements, and any amendments to those reports or statements. The Securities and Exchange Commission's ("SEC") website, www.sec.gov, contains reports, proxy and information statements, and other information regarding issuers that file electronically with the SEC. Materials filed or furnished by us with the SEC are also available at the SEC's Public Reference Room at 100 F Street, NE, Washington, D.C. 20549. Information on the operation of the Public Reference Room is available by calling (800) SEC-0330. The content on any website referred to in this Form 10-K is not incorporated by reference in this Form 10-K unless expressly noted.

Business Description:

Micron Technology Inc along with its subsidiaries provide memory and storage solutions. Its product portfolio consists of memory and storage technologies such as DRAM, NAND, NOR and 3D XPoint memory.

Address:	8000 South Federal Way P.O. Box 6 Boise, ID 83716-9632		
Telephone:	+1 208 368-4000	Website:	http://www.micron.com
Facsimile:	+1 208 368-4435	Email:	investorrelations@micron.com

Details

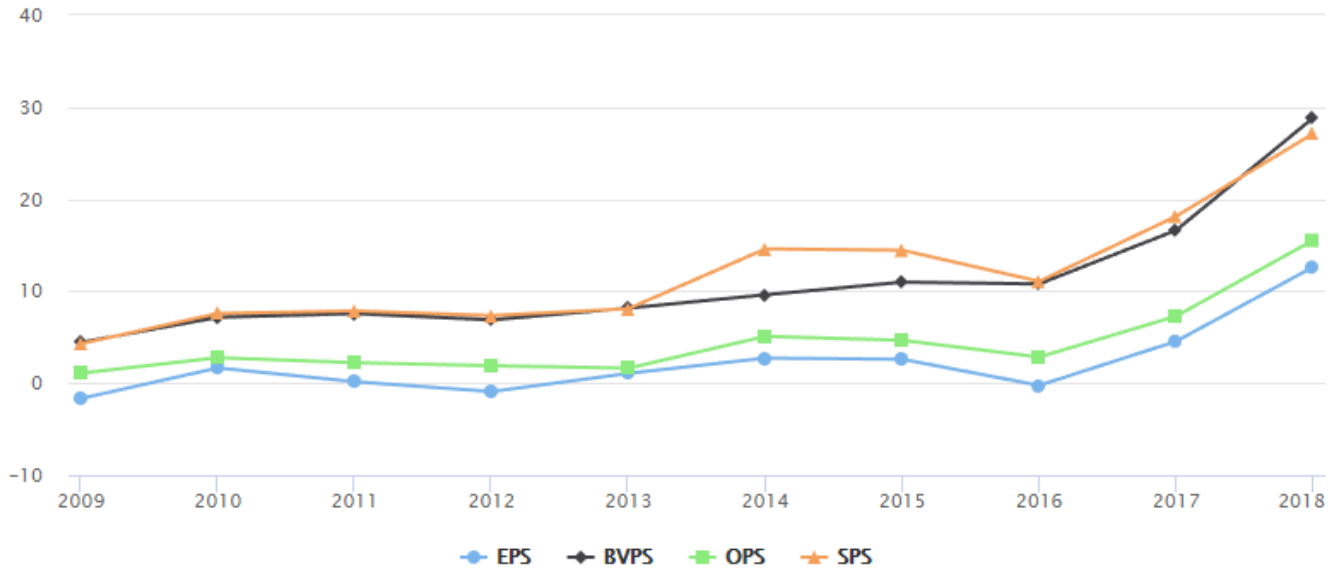
CEO:	Sanjay Mehrotra		
Employees:	36,000	Market Cap:	\$43.68 Bil.
Issue Type:	CS		
Auditor:	PricewaterhouseCoopers LLP		
Last Audit:	UQ		
Indices:	Dow Jones U.S. Semiconductors Index (^DJUSSC), Semiconductor Memory (^QM31169146), S&P 500 (^SPX)		

Industry Classifications

Sector:	Technology	CIK:	723125
Industry:	Semiconductors		
Sub-Industry:	Semiconductor Memory		
NAICS:	Semiconductor and Related Device Manufacturing(334413)		
SIC:	Semiconductors And Related Devices (3674)		

Moat

Moat



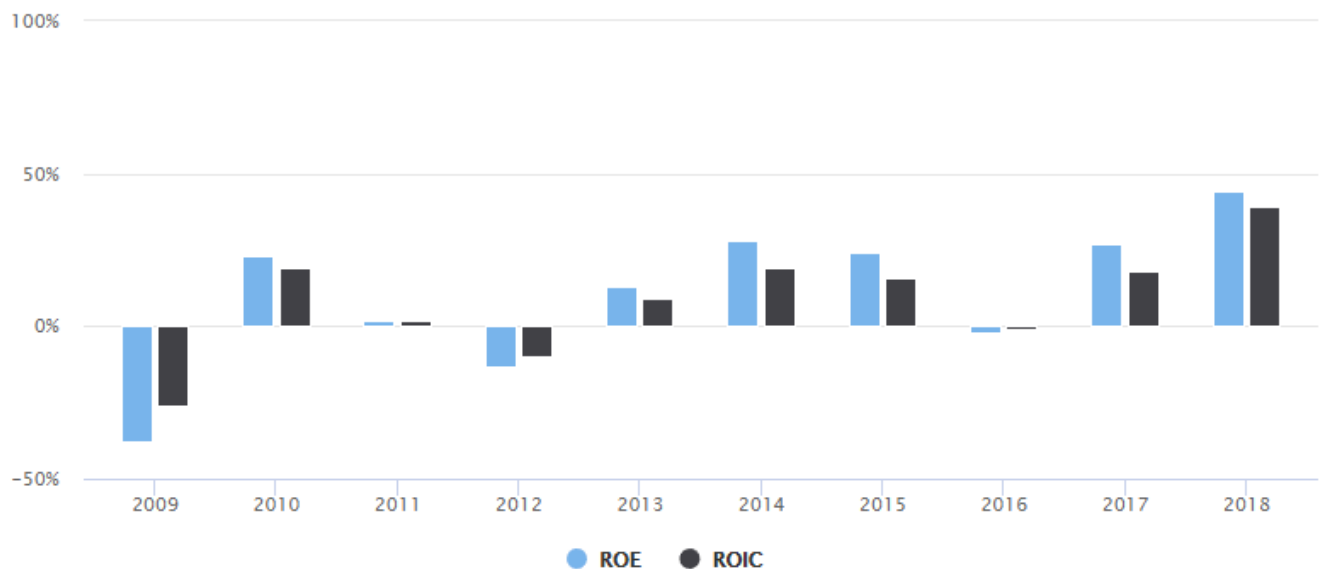
Seeking Alpha 9/14/18 Micron has a technology moat which will continue to grow after their joint venture with Intel concludes in 2019.

Consolidation in industry. Only 3 major players.

Management

One way to assess the quality of a company's management is to review its performance in efficiently employing capital. An efficient use of investment capital will usually produce consistent or growing return on equity (ROE) and return on invested capital (ROIC). Review the numbers and charts to see how management's effectiveness has been in the near past.

Management





December 6, 2018

Fellow Micron Shareholders,

Our fiscal 2018 was a year of unprecedented success for Micron. We set a host of new company records, both in overall revenue and profitability, as well as sales growth across multiple business segments. Our \$30 billion in revenue made us the fourth-largest semiconductor company in the world. We achieved approximately 50% operating margin, with profits that were greater than all previous years in the company's history combined. Our FY18 results were fueled by healthy end market trends for our products coupled with our strong execution on cost reduction and mix improvements. We doubled our overall sales to data center customers and had very strong growth across all our end markets, including automotive where we achieved record revenue and maintained our leading position.

In DRAM, we made significant gains in the fast-growing cloud data center segment through close relationships with key customers. We further strengthened our DRAM product portfolio in FY18 with the introduction of market-leading graphics and low-power DRAM products. We grew DRAM bits per wafer faster than the industry and achieved crossover to leading 1X nanometer DRAM in the client and graphics markets. In NAND, we achieved top market share in SATA enterprise solid state drives (SSDs) and strengthened our position in managed NAND solutions, which are popular in smartphones. I am proud to say that we have the most efficient 3D NAND design and were the first major memory manufacturer to introduce four-bit-per-cell SSDs.

We continue to accelerate our pace of innovation with a strong focus on reducing our cost structure faster than our competitors. In 2019, we will continue to increase our mix of high-value solutions as we introduce higher-density server DRAM and leading-edge SSDs, and we expect to launch 3D XPoint™ memory solutions that will ramp throughout 2020.

We are also investing to ensure our manufacturing operations remain on a strong footing. We began construction or started operations of manufacturing expansions in facilities in Singapore, Japan, and Taiwan this fiscal year and also announced plans to expand our fab in Manassas, Virginia. These investments will provide space for continued technology transitions in DRAM and NAND wafer manufacturing, as well as increased scale of captive assembly and test operations. These projects will help ensure our manufacturing operations become more cost-effective and streamlined for serving our customers in the future.

At our analyst event this past spring, our leadership team outlined our vision for "The New Micron." We are transforming our company to address fast-growing markets. There is an increasing demand for memory and storage solutions that enable data to be processed and analyzed quickly to extract the most value. Micron stands at the forefront of massive opportunity across multiple end markets, from artificial intelligence and autonomous driving, to data centers and the internet of things. As I look at a bright long-term future driven by secular tailwinds across these diverse markets, I am focused on strengthening the New Micron in five key areas:

- **Elevate Micron's position as an industry leader:** As noted above, we have made good progress this year on technology and product innovation. We are investing in our future and are building partnerships across industry and academia to ensure we can create long-term, sustainable growth that serves as a launching pad for future industry leadership.
- **Drive a high-performance company culture:** We are focused on creating and scaling an environment that accelerates decision-making and rewards outstanding performance. We are building on our strong base with the addition of significant new talent in the past year, including accomplished leaders, top-tier engineers and the best of the next generation of graduates.
- **Drive ongoing improvements in our financial results and position:** The New Micron is financially stronger than ever, with record FY18 profitability, improving margins, and a clear plan to deliver \$9 billion in operating efficiencies from 2016 to 2021 (we are already two-thirds of the way there). We continue to drive shareholder value, not just

through performance but with thoughtful capital allocation that balances R&D investment with debt reduction and the \$10 billion stock buyback plan first presented at our analyst event this past spring.

- **Build Micron's brand globally:** Micron is an icon in the technology industry. As our solutions become more essential to new technology trends, we are increasing our brand visibility to align world perception to the value we are creating. This fall, we hosted Micron's first-ever Insight Summit, where we outlined our position in artificial intelligence and other new technologies. We are collaborating closely with leading global companies to drive today's most important technology innovations.
- **Improve the lives of people around the world:** We continue working to achieve our vision to transform how the world uses information to enrich life. The capabilities our memory and storage solutions enable will drive dramatic improvements in healthcare, business efficiency, and life worldwide.

Beyond these major strategic objectives, Micron is committed to being an essential partner to the communities in which we operate through team member volunteering and corporate giving by the Micron Foundation. In calendar 2018 to date, our sites have already donated more than 100,000 hours of volunteerism to local causes. We also emphasize corporate responsibility and sustainability in all our operations and encourage you to review our 2018 Sustainability Report. Micron is focused on cultivating a diverse and inclusive culture in which all of our team members feel valued for who they are, how they think, and how they solve problems. We released our inaugural Diversity and Inclusion FY18 Annual Report where you can find more information about how we are addressing this key priority.

Micron celebrated our 40th anniversary worldwide this October. The rate of change and advancement in our industry over this time period has been astounding. Over those four decades, Micron has contributed nearly 40,000 patents to the industry. The technology leaps we have helped create provided the foundation for today's digital economy. Billions of people are now connected digitally, enabling growth and applications that were unimaginable just a few decades ago. I am immensely proud to be part of this innovative and tenacious team. Looking ahead, I am energized by the increasing role for Micron in the data economy as we enable broader adoption of exciting new technologies. Thank you for your support as a Micron shareholder. I hope you share my excitement for the potential ahead of us.

Sincerely,



Sanjay Mehrotra
President and CEO



December 7, 2017

Fellow Micron Shareholders,

Fiscal 2017 was a remarkable year for Micron. Intense focus and execution enabled us to harness industry tailwinds and achieve unprecedented results. For the first time in our nearly 40-year history, we surpassed \$20 billion in revenue, grew net income to a record \$5.1 billion, and generated a record \$3.3 billion in free cash flow.

Micron's business is more diversified than ever. We realized double-digit revenue growth in each of our four business units: Compute & Networking, Embedded, Mobile, and Storage. Our performance reflects the breadth of both our solutions and our end markets. Demand for memory and storage extends well beyond personal computers and mobile phones. Our technologies play a critical role in enabling some of today's most advanced applications, including artificial intelligence, autonomous driving, and virtual reality.

In this letter, I will recap our fiscal 2017 achievements, share our perspectives on the current industry dynamics, and highlight how our priorities for fiscal 2018 position us to capitalize on opportunities created by the evolving industry environment.

Executing Against our Strategy

Micron achieved several key objectives in fiscal 2017, which significantly improved our position across multiple aspects of the business. We successfully ramped 20nm DRAM and 32-layer 3D NAND technologies into volume production, enabling us to improve our relative cost competitiveness. We continued to execute our technology roadmap by introducing next-generation 1X DRAM and 64-layer 3D NAND, and we made substantial progress in follow-on DRAM and NAND technologies.

We furthered our efforts to increase the percentage of our revenue going to high-value solutions. This was particularly true for our NAND flash portfolio. Sales of solid state drives ("SSDs") were more than double what we achieved the year prior. Sales of managed NAND solutions, which are NAND devices packaged with controller technology, nearly doubled year-over-year.

We made meaningful improvements in manufacturing scale and efficiency. We completed the acquisition of Inotera Memories, driving improved margins and free cash flow, and enhancing our operational efficiency through greater scale. We also established a DRAM Center of Excellence in Taiwan, which will co-locate back-end test, assembly and packaging alongside our existing front-end DRAM fabrication facilities. This strategy mirrors our NAND Center of Excellence in Singapore and will drive further operational cost efficiencies across our global manufacturing footprint.

Finally, we strengthened our balance sheet in fiscal 2017, growing cash, marketable investments, and restricted cash to \$6.2 billion, while retiring \$1.5 billion in high-yield debt. We have continued to make progress towards being net cash positive by the end of fiscal 2018, retiring additional high-yield debt in November. This progress provides a healthier financial foundation upon which to grow revenues and profits.

Industry Environment and Fiscal 2018 Objectives

Today, memory and storage technologies play a more critical role in networking, storage and computing architectures than they have at any time in the past. There is an unprecedented amount of data being created, stored, and processed in a wide range of end-market applications and these applications are made possible with memory and storage solutions like those Micron develops. This trend creates opportunities for Micron to build more strategic relationships with our key customers and help them extract the most value from our technologies. We believe these opportunities will continue to grow as rapid advances in data analytics lead to new use cases and applications that improve our everyday lives.

These broad demand drivers have positively contributed to the current business climate. We have seen an increase in unit demand and average memory content across nearly every end market - from smartphones to servers to vehicles. We also saw a healthy balance between demand and industry supply throughout fiscal 2017. We expect industry supply growth to remain measured for both memory and storage technologies, largely balancing with demand throughout fiscal 2018.

We have identified four key areas of focus for fiscal 2018, which we consider instrumental to our future success. The entire Micron team is committed to excel against these objectives:

- **Cost Effectiveness** - While market diversification and advanced solutions are introducing new opportunities for high-value solutions, base technology costs remain fundamental to profitability in this business. We are focused on accelerating new technology development and ramping quickly and efficiently into volume production, while also reducing back-end costs, cycle time and inventory.
- **Product Execution** - Bringing the right solution at the right time ensures we can deliver the most value to our customers. Our engineering teams are adopting new methodologies to achieve the shortest possible customer qualification of our newest technology solutions.
- **Portfolio Management** - To prioritize revenue growth, profits and ROI, we will orient our product portfolio towards differentiated, high-value solutions. Doing so will also enable us to better respond to customer requirements and market trends.
- **Culture** - Hiring and keeping the industry's best talent will be a critical part of executing our strategy. We are implementing programs that promote a culture of achievement and success, which I believe will make Micron a global leader in competitive talent markets.

Our technologies are having a profound impact on the world around us. I trust that you share my enthusiasm for the significant opportunities that lie ahead for Micron, which are embodied in our vision statement: transforming how the world uses information to enrich life. Our efforts to positively impact the lives of people across the globe extend beyond the technologies we create. For example, through The Micron Foundation, we are enriching the communities in which we operate by partnering with educators in science, technology, engineering and math (STEM) to nurture tomorrow's technology pioneers. We published our second annual Sustainability Report in fiscal 2017, which reflects our commitment to supporting our global community as well as driving ethical and sustainable business practices.

As I close my first shareholder letter as Micron's CEO, I would like to acknowledge my predecessor, Mark Durcan. Micron is poised for great success in the years ahead, due in large part to his efforts to successfully position the Company through years of industry consolidation. I am proud to be leading this iconic company, particularly during a time when technology promises to create such dramatic change in each of our lives. Thank you for your continued support as a Micron shareholder. I look forward to sharing our progress with you over the year ahead.

Sincerely,



Sanjay Mehrotra

President and CEO

Margin of Safety

Earnings calls: 3/20, 6/20, 9/20, 12/20

Annual report: October

Very cyclical earnings. Cycle is about every 3-4 years. Long term growth great, but have to buy in a trough.

50% MOS (pre-loaded values, no customization)

MARGIN OF SAFETY CALCULATOR

Future Valuation

EPS TTM

12.13

?

EPS @ 1/27/2019 2:51 PM EST

Future Growth Rate:

6.30

?

Future EPS:

22.35

?

Future P/E:

12.60

?

Future Value:

281.61

?

MARR %:

15.00

?

MOS %:

50.00

?

FCF Per Share:

6.83

?

FCF Ratio %:

56.34

?

Green Zone Calculation:

Margin of Safety (MOS)

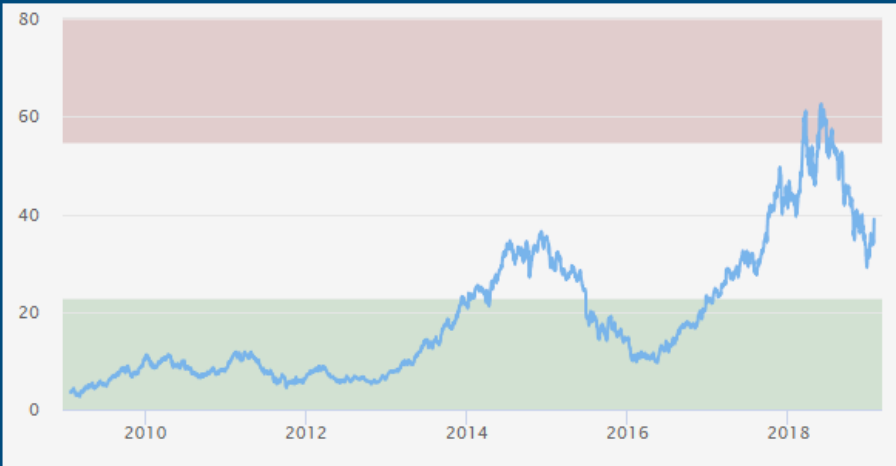
▼

8 yrs.

▼

RESET

SUBMIT



Stock Price:	\$38.96
P/E Ratio:	3.20
Low and High PE Ratio:	-- --
PEG:	0.32
Historical Growth Rate in BVPS:	18.0%
Price to Book:	1.29
Analyst Consensus Growth Rate:	10.00%
Earnings Yield:	32.36%

VALUATION RESULTS

STICKER PRICE:	\$45.54
MARGIN OF SAFETY PRICE:	\$22.77
MARGIN OF SAFETY YIELD:	43.92%
CURRENT PRICE FCF PAYBACK TIME:	5 Yrs.
FCF PAYBACK TIME PRICE at X Yrs:	\$59.87 at 8 Yrs.
RED ZONE PRICE:	\$54.64
GREEN ZONE PRICE:	\$22.77

50% MOS, customized



Free Cash Flow PBT

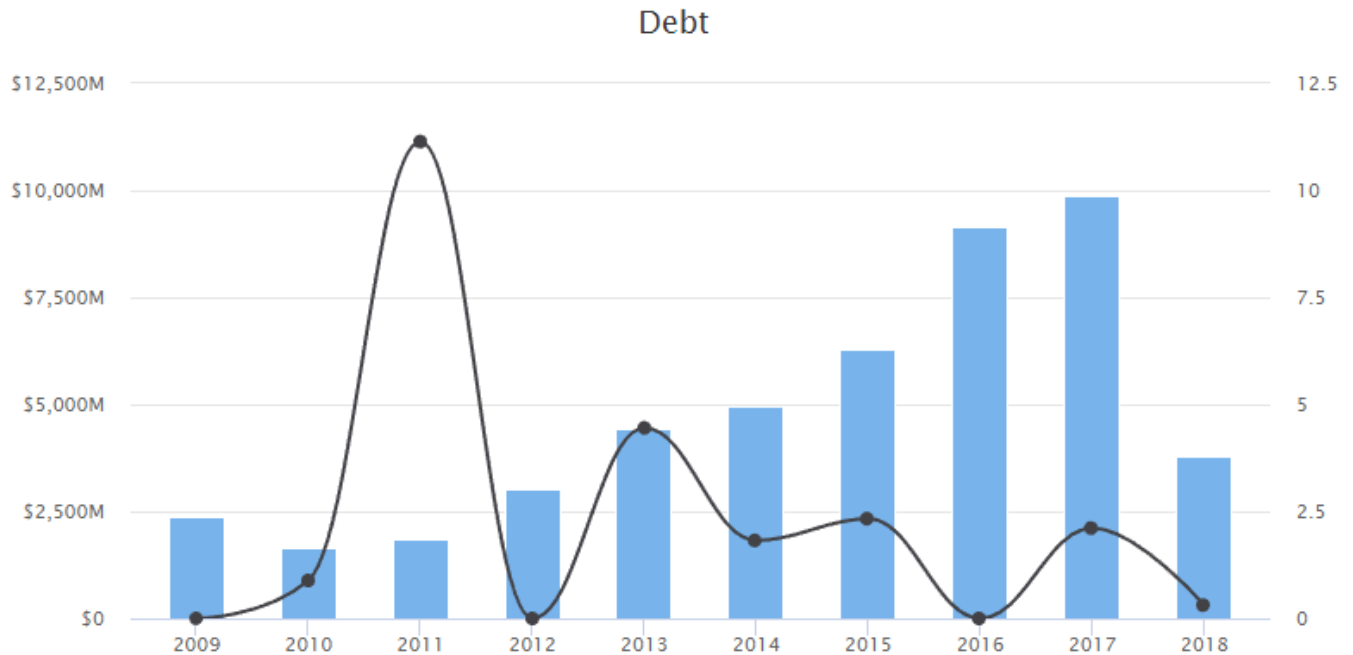
4 yrs (see excel sheet)

CAP RATE (using 2018 numbers only) uses owner earnings -

Pretax income 14,307
 Dep & Amort: 4,759
 Receivables: -1,734
 Payables: 549
 Minus maintenance Capex: -8,879
 \$9,002M = owner earnings
 1,120M = shares outstanding
 \$8.037/share of owner earnings
 Times 10 = \$80.37 10 Cap "on sale" price

The Three valuation methods

Future earnings 50% MOS:	\$34.19
8 yr Pay Back Time:	\$72.39
10 CAP (owner earnings)	\$80.37



Seeking Alpha 9/14/18

- Micron has announced an enormous stock buyback of over \$10 billion to start in their 2019 fiscal year, which has already started.
- Gross profit margins are steadily approaching 50%, which is among the highest in the industry.
- Micron has reduced the outstanding convertible securities, which will reduce the impact of shareholder dilution.
- Micron plans to return 50% of free cash flow, which would be \$4 billion based on 2018 performance.